Monday, May 4th

Sample space : set of all possible outcomes of a random experiment.

Examples:

1. we toss a coin once. $S = \{H, T\}$

2. we toss a coin twice.

$$S = \{HH, TT, HT, TH\}$$

3. we toss a coin until we observe a head

$$S = \{H, TH, TTH, TTTH, \ldots\}$$

Event: If S is countable, (discrete) any subset of S is called an event.

Simple event: It has only one outcome (sample point)

Example: we toss a coin twice

Event: we observe two heads $A = \{HH\}$

Compound event: It consists of multiple outcomes.

Example: toss a coin twice

Event: we observe at least one heads

$A = \{HT, TH, HH\}$

Wed. May 6th

① $P(\emptyset) = 0$

$\emptyset \cap S = \emptyset \quad , \quad S \cup \emptyset = S$

$$P(\cancel{S}) = P(S \cup \emptyset) = \cancel{P(S)} + P(\emptyset)$$

$$P(\emptyset) = 0$$

Alternatively

$\emptyset = \emptyset \cup \emptyset \cup \emptyset \ldots \qquad \emptyset \cap \emptyset = \emptyset$

$$P(\emptyset) = P(\emptyset \cup \emptyset \cup \emptyset \cup \ldots)$$

by Axiom 3

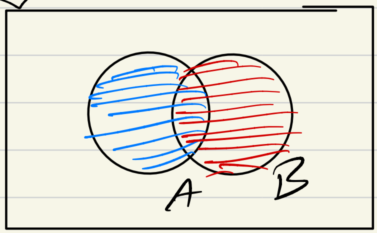
$$P(\emptyset) = P(\emptyset) + P(\emptyset) + P(\emptyset) + \ldots$$

by Axiom 1

$$P(\emptyset) = 0$$

② A and B are arbitrary events

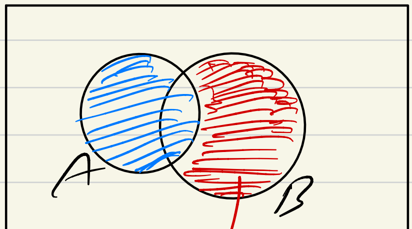
$$P(A\cup B)=P(A)+P(B)-P(A\cap B)$$



$$A\cup B = A\cup(B\cap A^c)$$

$$P(A\cup B)=P\big(A\cup(B\cap A^c)\big)$$

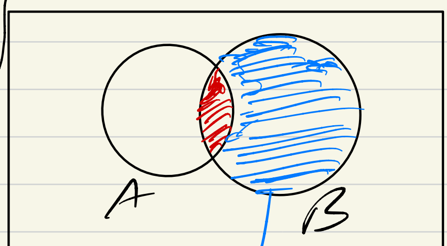
$$=P(A)+P(B\cap A^c)$$



$B\cap A^c = B-A$

$$B=(A\cap B)\cup(B\cap A^c)$$

$$P(B)=P(A\cap B)+P(B\cap A^c)$$



$B\cap A^c$

$$P(B)-P(A\cap B)=P(B\cap A^c)$$

$$\Rightarrow P(A\cup B)=P(A)+P(B)-P(A\cap B)$$

$\xrightarrow{}$ finite

Note: we can show that if $A_1, \dots, A_n$ are pairwise mutually exclusive ($A_i \cap A_j = \emptyset$, $i \neq j$)

Then $$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i)$$

To this end,

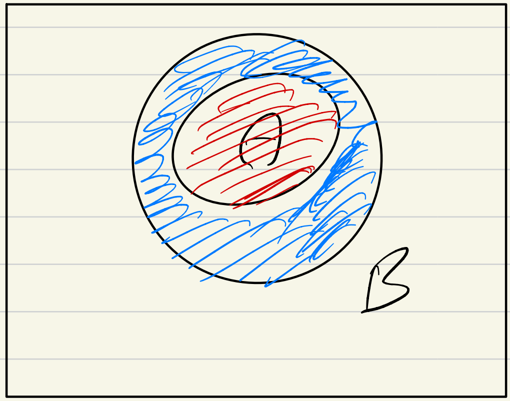
$A_{n+1} = A_{n+2} = \dots$ are all empty

$$P\left(\bigcup_{i=1}^{n} A_i\right) = P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \rightarrow \text{by Axiom 3}$$

$$= \sum_{i=1}^{n} P(A_i) + \underbrace{\sum_{i=n+1}^{\infty} P(A_i)}_{0}$$

③ if $A \subset B \Rightarrow P(A) \leq P(B)$

$B = A \cup (B \cap A^c)$



$P(B) = P(A \cup (B \cap A^c))$

$= P(A) + P(B \cap A^c)$

By A1) $\quad P(B) \geq P(A)$

4 Show that $p(A) \leq 1$

$$A \subset S$$

$$p(A) \leq p(S) = 1 \quad \text{by A2}$$

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5 $p(A^c) = 1 - p(A)$

$A \cap A^c = \phi$, $S = A \cup A^c$

$$1 = p(S) = p(A \cup A^c) = p(A) + p(A^c)$$

Axiom 2

$$p(A^c) = 1 - p(A)$$

Equilikely case

Suppose $S = \{x_1, \dots, x_m\}$ is a finite sample space a random experiment. Let $p_i = \frac{1}{m}$ for all $i = 1, \dots, m$, and for all subsets A of S, define

$$P(A) = \sum_{x_i \in A} \frac{1}{m} = \frac{\#(A)}{m}$$

$\#(A) \Rightarrow$ number of outcomes (elements) or sample points in A

then P is a probability on S, and it's referred to as the equilikely case.

Examples

Suppose we toss a fair coin twice

$S = \{HH, HT, TH, TT\}$

Find the probability that

1) they both are heads

$A = \{HH\}$ $\quad P(A) = \frac{1}{4}$

2) we observe at least one head

$B = \{HT, TH, HH\}$ $\quad P(B) = \frac{3}{4}$

Counting Rules:

mn Rules: A task consists of two parts. The part can be done in m ways, the second part can be done in n ways. The entire task can be done in mn ways.

Example: Travelling from Mississauga to Montreal via Toronto. (Bus, Train, Car)

Mississauga	Toronto	Montreal
	B_1	B_2
	T_1	T_2
	C	

(B_1, B_2), (B_1, T_2), (T_1, B_2), (T_1, T_2), (C, B_2), $(C, T_2$

$3 \times 2 = 6$

How to count number of ways to choose r objects from n objects.

without replacement

order is important (permutation)

$$\underbrace{\overset{1}{n} \times \overset{2}{(n-1)} \times (n-2) \times \cdots \overset{r}{(n-r+1)}}_{} = \frac{n!}{(n-r)!} = P^n_r$$

order is Not important (Combination)

$$\frac{n(n-1) \cdots (n-r+1)}{r \times (r-1) \times \cdots \times 2 \times 1} = \frac{n!}{(n-r)!\,r!} = C^n_r = \binom{n}{r}$$

with replacement

order is important

$$n \times n \times \cdots \times n = n^r$$

order is Not important

$$X_1 + X_2 + \cdots + X_n = r$$

$$\binom{n+r-1}{n-1}$$

Example:

Choose 2 letters from A, B, C

without replacements, order is important

$\{A, B\}, \{B, A\}$

$\{A, C\}, \{C, A\}$

$\{B, C\}, \{C, B\}$

$$P_2^3 = \frac{3!}{(3-2)!} = \frac{3 \times 2 \times 1}{1} = 6$$

$0! = 1$

$1! = 1$

$2! = 2 \times 1 = 2$

$3! = 3 \times 2 \times 1 = 6$

$$\frac{n! = n \times (n-1)(n-2) \times \cdots \times (n-r+1) \cancel{(n-r)} \cancel{(n-r-1)} \times \cdots \cancel{\times 2 \times 1}}{(n-r)! = \cancel{(n-r)} \cancel{(n-r-1)} \cdots \cancel{\times 2 \times 1}}$$

We choose two letters from A, B, C

without replacement. order is not important.

$\{A, B\}, \{A, C\}, \{B, C\}$

$$\frac{3!}{2!(3-2)!} = \frac{3 \times 2 \times 1}{2 \times 1} = 3$$

choose two letters from A, B, C

when sampling is with replacement, and

order is important

$\{A, B\}$, , $\{A, A\}$, $\{B, A\}$

$\{A, C\}$, $\{B, B\}$, $\{C, A\}$ $\qquad 3^2 = 9$

$\{B, C\}$, , $\{C, C\}$, $\{C, B\}$

$n = 3$, $\quad r = 2$ $\quad (0, 1, 1), (1, 0, 1)$

$X_1 + X_2 + X_3 = 2$ $\quad (1, 1, 0), (0, 0, 2)$

$(2, 0, 0), (0, 2, 0)$

$$\binom{3+2-1}{n-1} = \binom{4}{2} = \frac{4!}{2!\,2!} = \frac{4 \times 3}{2} = 6$$

Example 1.3.3

Find the probability that in a room with n people, at least two people have the same birthday.

$$P(A) = 1 - P(A^c)$$

$$= 1 - P(\text{All people have different Birthday})$$

$$= 1 - \frac{\overset{1}{365} \times 364 \times 363 \times \cdots (365-n+1)}{365^n}$$

$n = 20 \Rightarrow P(A) \approx 0.39$?

Example:

Suppose we have 4 stats books, 3 math books, 2 physic books. we randomly put them on a shelf. Find the probability that the books with the same subjects are next to each other.

$\#S = 9!$

$\#A = 3!(4! \times 3! \times 2!)$

Binomial Equation

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

For example : $n=2$

$$(x+y)^2 = x^2 + 2xy + y^2$$

$1 = \binom{2}{0}$ $\quad 2 = \binom{2}{1}$ $\quad 1 = \binom{2}{2}$

$\binom{n}{k} \Rightarrow$ Binomial coefficient.

$1 = \binom{3}{3}$

$$(x+y)^3 = \underset{1}{x^3} + 3xy^2 + 3x^2y + y^3$$

$1 = \binom{3}{0}$ $\quad 3 = \binom{3}{1}$ $\quad 3 = \binom{3}{2}$

Example:

Suppose we have 5 students, and want to create a union of 3 students. what is the number of ways to do that.

$$\binom{5}{3} = \frac{5!}{3!(5-3)!} = \frac{5!}{3!\,2!} = \frac{5\times \overset{2}{\cancel{4}}\times \cancel{3!}}{\cancel{3!}\,\cancel{2!}} = 10$$

In the previous example, what if the first student becomes the president, the second student becomes VP, and the third becomes the secretary.

order is important, but without replacement.

$$P_2^3 = \frac{3!}{(3-2)!} = 3! = 6.$$